

Mathematical Model for Predicting the Dissolved Oxygen Requirements of Steelhead (*Salmo gairdneri*) Embryos and Alevins in Hatchery Incubators

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ABSTRACT

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Routine metabolic rates ($\dot{V}O_2$), incipient limiting dissolved oxygen levels (P_c) and ambient oxygen levels resulting in one-half the routine metabolic rate (P_{50}) were determined for steelhead embryos and alevins incubated at constant temperatures of 6, 9, 12 and 15°C. Mathematical relationships were developed to predict $\dot{V}O_2$, P_c and P_{50} values for steelhead embryos and alevins incubated under both constant and variable temperature regimes. These relationships were incorporated into models that predict both available and required dissolved oxygen levels in various types of hatchery incubators. Practical applications of the incubation model are illustrated.

INTRODUCTION

Salmonid embryos and alevins are extremely sensitive to low levels of dissolved oxygen (Doudoroff and Shumway, 1970; European Inland Fisheries Advisory Commission, 1973; Davis, 1975; Alabaster and Lloyd, 1980). Some of the sublethal effects of hypoxia are retarded development (Garside, 1959, 1960), reduced vitelline circulation (Garside, 1959), delayed gill development (McDonald and McMahon, 1977), delayed or premature hatch depending on the past history of the embryo (Garside, 1966; Bailey et al., 1980), premature emergence (Bailey et al., 1980), reduced yolk conversion efficiency (Hamor and Garside, 1977), reduced size at hatch (Silver et al., 1963; Shumway et al., 1964), reduced size at emergence (Bailey et al., 1980) and teratogenesis

(Alderdice et al., 1958). Incipient limiting levels have not been determined precisely but appear to be relatively high. For example, Shumway et al. (1964) reported inhibition of growth in steelhead trout (*Salmo gairdneri*) embryos incubated at $8.1 \text{ mg O}_2 \text{ l}^{-1}$ at 10°C .

Several investigators have suggested that accurate response thresholds could be determined most easily by making use of the fact that salmonid embryos and alevins are metabolic regulators (Hayes et al., 1951; Wickett, 1954, 1975; Alderdice et al., 1958). The metabolic rate of embryos and alevins is independent of the ambient oxygen level at high oxygen concentrations. However, as oxygen levels decline, a point is reached below which metabolic rate becomes dependent on the ambient oxygen concentration. This point was termed the incipient limiting level by Fry (1957) and the critical oxygen level by Prosser and Brown (1962). Growth and survival will be optimal at oxygen concentrations above the critical level, providing other environmental variables are adequate. Critical dissolved oxygen levels are used in this paper to estimate the minimum dissolved oxygen requirements of steelhead embryos and alevins.

Although most hatchery managers recognize that adequate levels of dissolved oxygen are essential for successful hatchery operations, it is not always practical to monitor oxygen levels within incubators. Remedial action is usually not taken until obvious symptoms appear, by which time irreversible injury may have occurred. However, incubation systems are essentially "closed" with regard to oxygen supply and it is possible to predict oxygen levels in incubators if metabolic rate, number of individuals, inflow oxygen level, flow rate, reaeration coefficient and temperature are known (West et al., 1982). This paper is one of two on the oxygen requirements of steelhead embryos and alevins. In the first paper, published elsewhere (Rombough, 1986), were presented the biological data, in the present paper, mathematical relationships are given predicting dissolved oxygen levels in incubators, and practical applications of the ensuing model for eggs incubating in a hatchery environment are discussed.

MATERIALS AND METHODS

Gametes of the steelhead trout (*Salmo gairdneri*) were obtained from fish returning to spawn in the Big Qualicum River, Vancouver Island, B.C. Eggs were pooled from two females and fertilized by the dry method with the pooled milt of two males. Water-hardened eggs had a mean wet weight ($\pm \text{SD}$, $n=20$) of $147.0 \pm 7.0 \text{ mg}$ and an average diameter of 6.15 mm. Eggs and alevins were incubated at nominal temperatures of 6, 9, 12, and 15°C from fertilization to near complete yolk absorption in vertically stacked incubation trays (Heath Techna Corp., Seattle, WA). Initial loading densities were approximately 2500 eggs per tray. Water flows were maintained at 12 l min^{-1} using dechlorinated water from the Nanaimo municipal supply (total hardness 12 mg l^{-1} as CaCO_3 ; pH 7.2). Actual water temperatures were monitored daily and averaged ($\pm \text{SD}$)

6.0 ± 0.4 , 9.1 ± 0.4 , 12.0 ± 0.4 and $15.1 \pm 0.3^\circ\text{C}$ throughout incubation. Dissolved oxygen levels were monitored periodically and always exceeded 90% of air saturation.

Routine metabolic rates ($\dot{V}\text{O}_2$), critical dissolved oxygen levels (P_c) and oxygen levels at which metabolic rate was one-half the routine rate (P_{50}) were determined at 10–12 approximately evenly spaced intervals between fertilization and complete yolk absorption at each temperature. Intervals averaged 12.8, 7.2, 5.0 and 3.8 days, respectively, at 6, 9, 12 and 15°C . Metabolic rates, P_c and P_{50} values were determined by monitoring oxygen consumption in three closed system, continuous flow respirometers. An identical fourth respirometer served as a blank control. Respirometers were constructed of acrylic and stainless steel and had total volumes of between 70.25 and 71.88 ml. Dissolved oxygen levels in the respirometers were monitored continuously using polarographic oxygen electrodes (Orbisphere model 2104.01), a multichannel oxygen meter (Orbisphere model 2710) and strip chart recorders. A more detailed description of the respirometer system is given by Rombough (1987).

Respirometers were operated routinely in the "closed" mode. In most instances, the initial level was 100% of air saturation. However, oxygen levels were elevated to between 110 and 160% of air saturation for tests with late-eyed embryos and newly hatched alevins to insure that initial oxygen levels were above P_c . Between 30 and 300 individuals, depending on temperature and stage of development, were placed in each respirometer. The apparent water velocity in the respirometer was adjusted to 500 cm h^{-1} (corresponding to a flow rate of about 12 l min^{-1} in a Heath tray) and the respirometer assembly covered with heavy black plastic. Embryos and alevins were allowed 30 min in virtual darkness to accommodate to the respirometer. Respirometers were "closed" and dissolved oxygen levels in the respirometers were permitted to drop to $1\text{--}2\text{ mg l}^{-1}$ before the systems were reopened. The duration of the tests ranged from 50 min to 5 h: most tests were completed within 1–1.5 h.

Embryos and alevins were removed from the respirometers at the end of the tests and five individuals were preserved in 5% neutral formalin. Developmental stages were ascertained using the system of Vernier (1969). Samples were fixed for at least 21 days, dissected into tissue and yolk components, and weighed to the nearest 0.1 mg. Tissue and yolk were dried at 60°C for 48 h and dry weights determined.

Routine metabolic rates, critical dissolved oxygen levels and P_{50} values were obtained from the polarographic records. Metabolic rates were determined by taking the tangent to the line of declining oxygen concentration as a function of time and applying an appropriate scaling factor after correcting for control uptake. Apparent oxygen uptake was rapid initially because of pressure changes associated with closing the system. After about 5–15 min, electrodes stabilized and oxygen levels in respirometers declined at a constant rate. Typically, the rate of change remained constant while ambient levels dropped $3\text{--}6\text{ mg l}^{-1}$.

The rate of oxygen consumption during this period was taken to be the routine metabolic rate. Eventually, the polarographic record began to flatten out. The point at which it first deviated perceptibly from a straight line was deemed to represent the critical oxygen level. The ambient level at which the apparent rate of uptake was one-half the routine rate was taken to be the P_{50} value. Analysis of the polarographic record and comparison of metabolic rates determined using the respirometers in "closed" and "open" modes are discussed in more detail by Rombough (1987).

Times to maximum alevin wet weight, maximum tissue weight and ultimate maximum metabolic rate at the various experimental temperatures were estimated by fitting the data to a polynomial of the type

$$y = a + bx + cx^2$$

where y is alevin wet weight, tissue wet weight or metabolic rate and x is the number of days postfertilization, and solving for the maximum (Rombough, 1985). Data relating times to completion of epiboly (Vernier, 1969, stage 17), the "well-eyed" stage (stage 24), 50% hatch, ultimate maximum metabolic rate and maximum alevin wet weight to incubation temperature were fitted to linear, exponential, semilogarithmic and parabolic models using the Minitab regression computer program (Ryan et al., 1982). Equations of best fit relating tissue wet weight, routine metabolic rate, critical dissolved oxygen level and P_{50} values to incubation period and temperature were calculated using the Minitab multiple regression program (Ryan et al., 1982).

RESULTS

Embryonic mortality (fertilization to hatch) was 4.0, 2.6, 3.4 and 15.0%, respectively, at 6, 9, 12 and 15°C. Alevin mortality was less than 5% at all temperatures.

Growth curves for steelhead embryos and alevins are displayed in Fig. 1. Tissue growth during the period between completion of epiboly (stage 17) and maximum tissue weight is described by the equation

$$\ln \text{TWW} = -12.718 + 2.685 \ln d + 3.050 \ln T \quad (R^2 = 98.5\%, n = 34) \quad (1)$$

where TWW is tissue wet weight (mg), d is number of days postfertilization and T is incubation temperature (°C). Actual times to completion of epiboly (stage 17), the "well-eyed" stage (stage 24), 50% hatch, maximum alevin wet weight and maximum tissue wet weights for steelhead incubated at constant temperatures are given in Table 1. The equations of best fit predicting time to completion of epiboly (Eqn. 2), time to the "well-eyed" stage (Eqn. 3), time to 50% hatch (Eqn. 4), time to maximum metabolic rate (Eqn. 5) and time to maximum alevin wet weight (Eqn. 6) at constant incubation temperatures are given in Table 2.

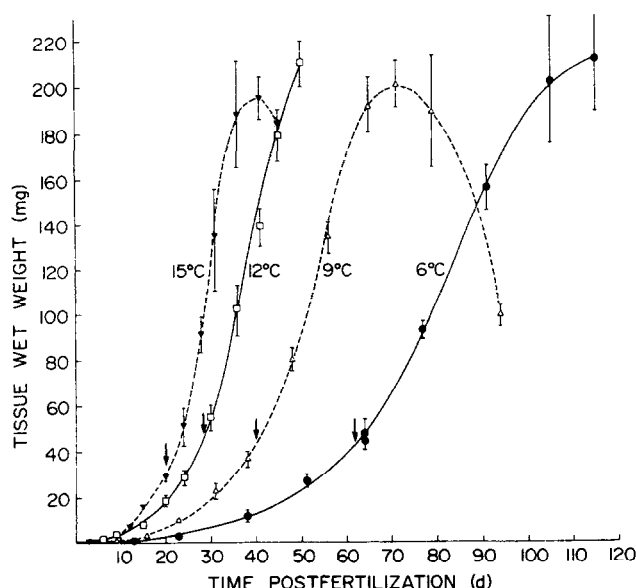


Fig. 1. Relationships between tissue wet weight and incubation period for steelhead embryos and alevins incubated at 6, 9, 12 and 15°C. Mean values and 95% confidence limits are plotted. Arrows indicate 50% hatch.

The routine metabolic rate of steelhead increased 200–500 fold, depending on temperature, during development. Metabolic rates were initially low but increased rapidly once formation of the embryonic axis was complete (stage

TABLE 1

Times to various developmental stages for steelhead embryos and alevins incubated at constant temperatures; stages were identified using the system of Vernier (1969)

Temperature (°C)	Time to completion of epiboly (stage 17) (d)	Time to "well-eyed" (stage 24) (d)	Time to 50% hatch (d)	Time to maximum alevin wet weight (d)	Time to maximum tissue wet weight (d)
4.5*	22.0	39.9	85.0	—	—
6.0	—	—	61.7	110.5	> 115.0
7.0*	12.1	25.8	50.8	—	—
8.5*	10.1	20.8	45.0	—	—
9.1	—	—	40.0	68.6	72.7
12.0	—	—	27.9	> 50.0	> 50.0
14.0*	5.7	10.5	22.0	—	—
15.1	—	—	20.0	38.0	39.7

*F. Velsen, Pacific Biological Station, personal communication, 1985.

TABLE 2

Minimum variance estimates of the parameters in the equations of best fit relating time to various developmental stages (d , days) and constant incubation temperature (T , °C). Linear, exponential, semilogarithmic and parabolic models were tested. Stages are after Vernier (1969)

Time to stage (days)	Equation of best fit	R^2 (%)	n
Completion of epiboly (stage 17)	$\ln d = 4.85 - 1.19 \ln T$ (2)	99.5	4
"well-eyed" (stage 24)	$\ln d = 4.25 - 0.138 T$ (3)	98.8	4
50% hatch	$\ln d = 6.27 - 1.19 \ln T$ (4)	99.0	8
Maximum metabolic rate	$\ln d = 6.67 - 1.13 \ln T$ (5)	98.9	4
Maximum alevin wet weight	$\ln d = 5.35 - 0.115 T$ (6)	98.6	3

17). Metabolic rates reached a maximum shortly before maximum alevin wet weight and then declined (Fig. 2). Maximum metabolic rates prior to completion of epiboly, ultimate maximum metabolic rates and times to ultimate maximum metabolic rates at the various experimental temperatures are given in Table 3. Equations providing the best fit to the experimental data relating

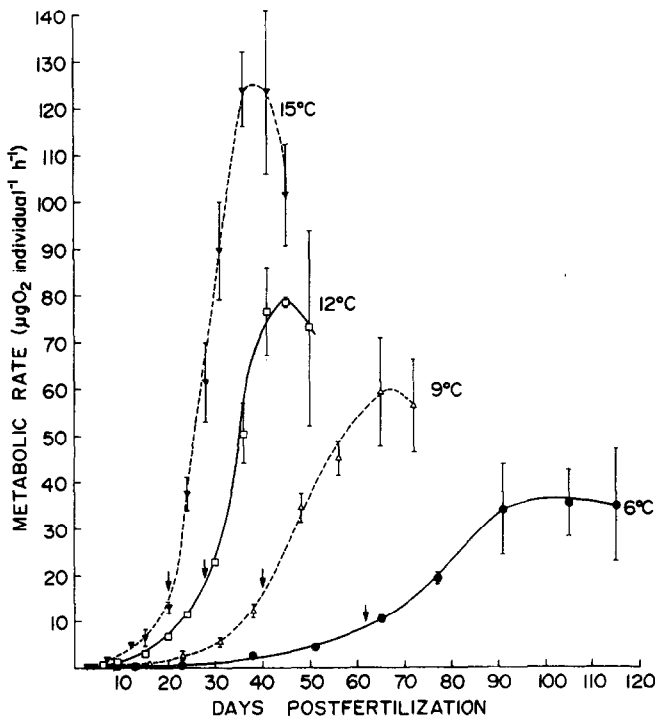


Fig. 2. Relationships between routine metabolic rate and incubation period for steelhead embryos and alevins incubated at 6, 9, 12 and 15°C. Mean values and 95% confidence limits are plotted. Arrows indicate 50% hatch.

TABLE 3

Maximum metabolic rate prior to completion of epiboly (stage 17; Vernier, 1969), ultimate maximum metabolic rate and time to ultimate maximum metabolic rate for steelhead embryos and alevins incubated at constant temperatures

Temperature (°C)	Maximum metabolic rate prior to completion of epiboly ($\mu\text{g O}_2 \text{ h}^{-1}$)	Ultimate maximum alevin metabolic rate ($\mu\text{g O}_2 \text{ h}^{-1}$)	Time to ultimate maximum alevin metabolic rate (d)
6.0	0.17	37.0	105.1
9.1	0.28	60.0	67.1
12.0	0.60	79.0	44.9
15.1	0.96	125.0	38.5

maximum metabolic rates prior to completion of epiboly to temperature (Eqn. 7), routine metabolic rate during the period between blastopore closure and ultimate maximum metabolic rate to incubation period and temperature (Eqn. 8), and ultimate maximum metabolic rate to incubation temperature (Eqn. 9) are given in Table 4.

Critical oxygen values increased from low levels at fertilization to a maximum immediately prior to hatch. Levels dropped $2\text{--}3 \text{ mg l}^{-1}$ at hatch and gradually declined to stable levels about mid-way through the alevin stage (Fig. 3). P_{50} values followed a pattern similar to that observed for P_c during embryonic development but unlike P_c did not vary significantly during the alevin stage.

TABLE 4

Minimum variance estimates of the parameters in the equations of best fit relating routine metabolic rate ($\dot{V}\text{O}_2$; $\mu\text{g individual}^{-1} \text{ h}^{-1}$) of steelhead incubated at constant temperature to incubation period (d , days postfertilization) and incubation temperature (T , °C) during periods (1) prior to completion of epiboly (Eqn. 7), (2) between completion of epiboly and ultimate maximum metabolic rate (Eqn. 8), and (3) between ultimate maximum metabolic rate and maximum alevin wet weight (Eqn. 9)

Incubation period	Equation of best fit		R^2 (%)	n
Fertilization to completion of epiboly	$\ln \dot{V}\text{O}_2 = -2.977 + 0.1969 T$	(7)	98.3	4
Completion of epiboly to ultimate maximum metabolic rate	$\ln \dot{V}\text{O}_2 = -18.139 + 2.950 \ln d + 4.549 \ln T$	(8)	97.9	28
Ultimate maximum metabolic rate to maximum alevin wet weight	$\ln \dot{V}\text{O}_2 = 2.852 + 0.1302 T$	(9)	98.9	4

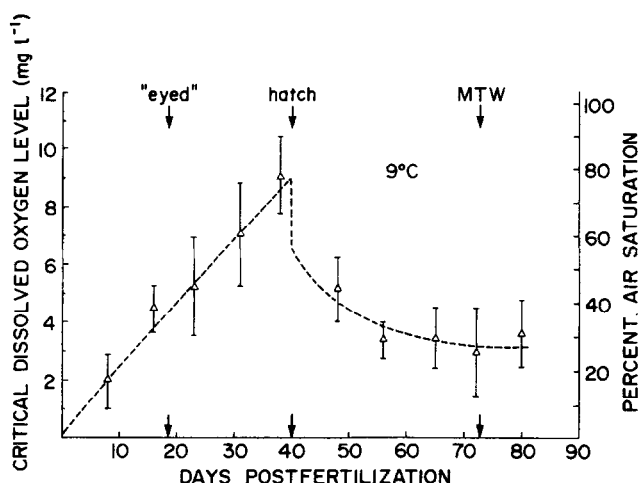


Fig. 3. Critical dissolved oxygen levels for steelhead embryos and alevins incubated at a constant temperature of 9°C. Mean values and 95% confidence limits are plotted.

Estimated maximum P_c and P_{50} values for embryos, estimated minimum P_c values for alevins, and actual mean P_{50} values for alevins are listed in Table 5. Critical oxygen levels and P_{50} values (mg l^{-1}) for embryos as functions of incubation period (days postfertilization) and temperature ($^{\circ}\text{C}$) are estimated by Eqns. 10 and 11, respectively (Table 6). Similarly, critical levels (mg l^{-1}) for alevins as functions of incubation period (days posthatch) and temperature ($^{\circ}\text{C}$) are predicted by Eqn. 12 (Table 6). P_{50} values are described as a function of incubation temperature in Eqn. 13 (Table 6). Equations predicting maximum P_c (Eqn. 14) and P_{50} (Eqn. 15) levels for embryos as a function of temperature are given in Table 7. Minimum P_c values for alevins are predicted by Eqn. 16 (Table 7).

TABLE 5

Estimated maximum critical oxygen levels and P_{50} values for embryos and minimum critical oxygen levels and mean P_{50} values for alevins of steelhead incubated at constant temperatures

Temperature ($^{\circ}\text{C}$)	Estimated maximum P_c value for embryos (mg l^{-1})	Estimated maximum P_{50} value for embryos (mg l^{-1})	Estimated minimum P_c value for alevins (mg l^{-1})	Mean P_{50} value for alevins (mg l^{-1})
6.0	7.6	2.6	2.5	1.2
9.1	8.7	3.1	3.1	1.8
12.0	9.5	3.6	3.7	1.9
15.1	10.2	4.0	4.5	2.4

TABLE 6

Minimum variance estimates of the parameters in the equations of best fit relating critical dissolved oxygen levels (P_c , mg l^{-1}) and P_{50} values (P_{50} , mg l^{-1}) for steelhead incubated at constant temperature to incubation period (dpf, days postfertilization for embryos; dph, days posthatch for alevins) and temperature (T , $^{\circ}\text{C}$)

Incubation period	Equation of best fit	R^2 (%)	n
Critical level for embryos	$\ln P_c = -4.50 + 0.949 \ln \text{dpf} + 1.45 \ln T$ (10)	97.1	20
P_{50} value for embryos	$\ln P_{50} = -8.11 + 1.31 \ln \text{dpf} + 2.03 \ln T$ (11)	79.5	19
Critical level for alevins	$P_c = 5.66 - 1.04 \ln \text{dph} + 0.131 T$ (12)	79.5	20
P_{50} values for alevins	$P_{50} = 4.81 + 0.127 T$ (13)	92.8	20

DISCUSSION

Reliable estimates of routine metabolic rates are required for accurate prediction of oxygen levels within incubators. Calculation of energy budgets for the period between fertilization and 90% yolk utilization indicated that the metabolic rates determined for steelhead embryos and alevins were close to the actual routine metabolic rates during this period (Rombough, 1987). It was not possible to use a single equation to predict routine metabolic rates accurately throughout development. Consequently, the total incubation period was divided into segments, and different equations were used to predict rates of oxygen consumption during each segment. The segments chosen were: (1) fertilization to completion of epiboly; (2) completion of epiboly to ultimate maximum metabolic rate; and (3) ultimate maximum metabolic rate to maximum alevin wet weight (Fig. 4). Routine metabolic rate was estimated only once prior to completion of epiboly at each temperature. The relationship

TABLE 7

Minimum variance estimates of the parameters in the equations of best fit relating maximum P_c and P_{50} values for embryos and minimum P_c values for alevins (mg l^{-1}) to incubation temperature ($^{\circ}\text{C}$)

Incubation period	Equation of best fit	R^2 (%)	n
Maximum P_c for embryos	$P_c = 2.45 + 2.81 \ln T$ (14)	99.8	4
Maximum P_{50} for embryos	$P_{50} = 1.68 + 0.156 T$ (15)	99.5	4
Minimum P_c for alevins	$\ln P_c = 0.501 + 0.688 T$ (16)	99.9	4

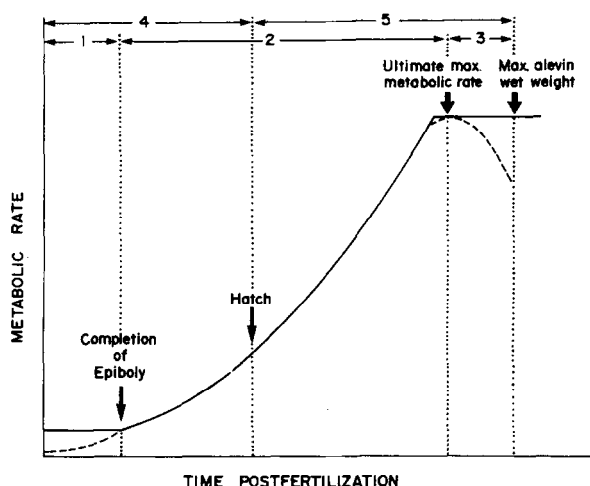


Fig. 4. Schematic diagram showing the segments into which the total incubation period is divided for the purposes of modelling oxygen uptake rates, P_c and P_{50} values. Equations are developed to predict routine metabolic rate during the periods (1) between fertilization and completion of epiboly, (2) between completion of epiboly and ultimate maximum metabolic rate, and (3) between ultimate maximum metabolic rate and maximum alevin wet weight. P_c and P_{50} are estimated separately for the periods (4) between fertilization and hatch and (5) between hatch and maximum alevin wet weight.

between this value and temperature described by Eqn. 7 was used to predict rates of oxygen consumption during segment (1). Routine metabolic rates during the period between completion of epiboly and ultimate maximum metabolic rate, segment (2), are predicted accurately by the parabolic relationship between rate of oxygen consumption and time and temperature given in Eqn. 8. Routine metabolic rates declined after reaching a maximum shortly before maximum alevin wet weight (Fig. 2). However, the relationship between ultimate maximum metabolic rate and temperature described by Eqn. 9 was used to estimate oxygen uptake throughout segment (3) because alevins may be relatively active during this period, and the use of the maximum metabolic rate provides a safety margin to compensate for any increased oxygen uptake associated with activity.

Critical dissolved oxygen levels and P_{50} values, similarly, could not be predicted for embryos and alevins using the same equations. P_c and P_{50} values were predicted using Eqns. 10 and 11 for the period before hatch [segment (4), Fig. 4] and by Eqns. 12 and 13 for the period after hatch [segment (5), Fig. 4].

Metabolic rates, P_c and P_{50} values were determined for steelhead at constant temperatures. However, embryos and alevins are incubated under a variable thermal regime at most hatcheries. Predictive models must take this into account if they are to be of practical use. One of the simplest ways to account

TABLE 8

Equations predicting the length of time required to reach various developmental stages for steelhead incubated under a variable temperature regime. D is the number of days required to reach a particular stage, ATU is accumulated thermal units from fertilization ($^{\circ}\text{C}\cdot\text{d}$) and d is the actual number of days from fertilization

Stage	Equation	
Completion of epiboly (stage 17)	$\ln D = 4.85 - 1.19 \ln (\text{ATU } d^{-1})$	(17)
"well eyed" (stage 24)	$\ln D = 4.25 - 0.138 \ln (\text{ATU } d^{-1})$	(18)
50% hatch	$\ln D = 6.27 - 1.19 \ln (\text{ATU } d^{-1})$	(19)
Maximum metabolic rate	$\ln D = 6.67 - 1.13 \ln (\text{ATU } d^{-1})$	(20)
Maximum alevin wet weight	$\ln D = 5.35 - 0.115 (\text{ATU } d^{-1})$	(21)

for changing temperatures during development is to make use of the thermal sums hypothesis. The hypothesis states that the product of the length of time required to reach a particular stage of development and the mean incubation temperature during the period is a constant (Wallich, 1901). The constant is normally expressed as accumulated thermal units (ATU), that is, the sum of the mean daily temperatures ($^{\circ}\text{C}\cdot\text{d}$). The thermal sums model is known to be somewhat inaccurate, particularly near the upper and lower limits of thermal tolerance, but is widely used in hatcheries because of its simplicity. Alderdice and Velsen (1978) incorporated elements of this hypothesis into more sophisticated models to predict hatching times for chinook salmon (*Oncorhynchus tshawytscha*) incubated under variable temperature regimes. They reported no significant difference in time to hatch for constant and variable temperature incubations if the mean temperature ($\bar{T} = \text{ATU } d^{-1}$) was used as the predictor. Rombough (1985), similarly, observed no significant difference in times to maximum alevin wet weight for chinook incubated at constant and variable temperatures when mean temperature ($\bar{T} = \text{ATU } d^{-1}$) was used in a semilogarithmic model to express thermal history. This approach was used to predict times to various developmental stages for steelhead incubated under fluctuating temperature regimes. For example, Eqn. 2, predicting time to completion of epiboly at constant temperature, becomes

$$\ln D = 4.85 - 1.19 \ln (\text{ATU } d^{-1}) \quad (17)$$

where D is days to completion of epiboly, ATU is accumulated thermal units from fertilization ($^{\circ}\text{C}\cdot\text{d}$) and d is actual days after fertilization. Equations predicting times to eyeing (Eqn. 18), 50% hatch (Eqn. 19), maximum metabolic rate (Eqn. 20) and maximum alevin wet weight (Eqn. 21) were similarly modified and are given in Table 8.

Equations predicting metabolic rate, P_c and P_{50} at constant temperatures also must be modified to account for the variable temperature regime of hatch-

TABLE 9

Equations predicting metabolic rates of steelhead embryos and alevins incubated under a variable thermal regime. $\dot{V}O_2$ is routine metabolic rate ($\mu\text{g O}_2 \text{ h}^{-1}$), ATU is accumulated thermal units from fertilization ($^{\circ}\text{C}\cdot\text{d}$) and T' is current ambient water temperature ($^{\circ}\text{C}$)

Incubation period	Equation	
Fertilization to completion of epiboly	$\ln \dot{V}O_2 = -2.977 + 0.1969 T'$	(22)
Completion of epiboly to ultimate maximum metabolic rate	$\ln \dot{V}O_2 = -18.139 + 2.950 \ln (\text{ATU } T'^{-1}) + 4.549 \ln T'$	(23)
Ultimate metabolic rate to maximum alevin wet weight	$\ln \dot{V}O_2 = 2.852 + 0.1302 T'$	(24)

eries. Metabolic rates prior to completion of epiboly and after ultimate maximum metabolic rate, described by Eqns. 7 and 9, respectively, were assumed to be dependent only on the ambient temperature and not on stage of development so do not require modification (Eqns. 22 and 24, Table 9). However, metabolic rate at constant temperature was dependent on both incubation period (d) and temperature (T) during the period between completion of epiboly and maximum metabolic rate (Eqn. 8) and must be modified. The simplest way to do this is to replace incubation period (d) in Eqn. 8 by an expression representing the length of time that would have been required to reach the same stage of development (ATUs) at the current ambient temperature. Equation 8 thus becomes

$$\ln \dot{V}O_2 = -18.139 + 2.950 \ln (\text{ATU } T'^{-1}) + 4.549 \ln T' \quad (23)$$

where $\dot{V}O_2$ is the routine metabolic rate ($\mu\text{g O}_2 \text{ h}^{-1}$), ATU is accumulated thermal units from fertilization ($^{\circ}\text{C}\cdot\text{d}$) and T' is the current ambient temperature ($^{\circ}\text{C}$) (Table 9). Actual mean metabolic rates at 6, 9, 12 and 15°C and curves predicted for those temperatures using Eqns. 22, 23 and 24 are shown as a function of accumulated thermal units after fertilization in Fig. 5.

Equations predicting P_c and P_{50} for embryos and alevins were modified similarly for application to variable temperature regimes (Table 10). The regression predicting critical levels for embryos thus becomes

$$\ln P_c = -4.50 + 0.949 \ln (\text{ATU } T'^{-1}) + 1.45 \ln T' \quad (25)$$

where P_c is the critical level (mg l^{-1}), ATU is accumulated thermal units from fertilization ($^{\circ}\text{C}\cdot\text{d}$) and T' is the current ambient temperature ($^{\circ}\text{C}$). Actual

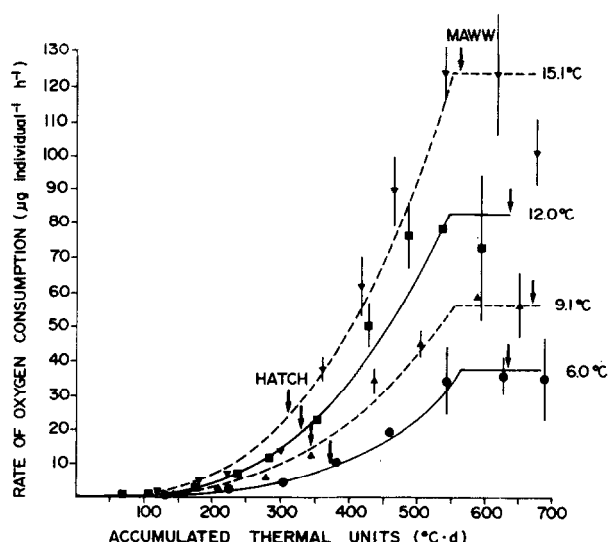


Fig. 5. Routine metabolic rates of steelhead incubated at constant temperatures of 6, 9, 12, and 15°C. Mean values and 95% confidence limits are plotted as functions of accumulated thermal units ($^{\circ}\text{C}\cdot\text{d}$) from fertilization. Curves were fitted using Eqns. 22, 23 and 24 relating routine metabolic rate to incubation period expressed in accumulated thermal units and ambient temperature for segments 1, 2 and 3, respectively (see text). Arrows indicate accumulated thermal units to 50% hatch and maximum alevin wet weight (MAWW).

P_c values for embryos incubated at 6, 9, 12 and 15°C and the curves for those temperatures predicted using Eqn. 25 are displayed as a function of accumulated thermal units from fertilization in Fig. 6. The equivalent regression predicting critical levels for alevins under variable thermal regimes is

$$P_c = 1.04 \ln (\text{ATU}_a T'^{-1}) + 0.131 T' \quad (27)$$

where P_c is the critical level (mg l^{-1}), ATU_a is accumulated thermal units from hatch ($^{\circ}\text{C}\cdot\text{d}$) and T' is current ambient temperature ($^{\circ}\text{C}$). Actual P_c

TABLE 10

Equations predicting P_c and P_{50} values for steelhead embryos and alevins incubated under variable temperature regimes. P_c and P_{50} are in mg l^{-1} , ATU is accumulated thermal units from fertilization ($^{\circ}\text{C}\cdot\text{d}$), ATU_a is accumulated thermal units from hatch ($^{\circ}\text{C}\cdot\text{d}$) and T' is current ambient temperature ($^{\circ}\text{C}$)

Incubation period	Equation	
Critical level for embryos	$\ln P_c = -4.50 + 0.949 \ln (\text{ATU } T'^{-1}) + 1.45 \ln T'$	(25)
P_{50} value for embryos	$\ln P_{50} = -8.11 + 1.31 \ln (\text{ATU } T'^{-1}) + 2.03 \ln T'$	(26)
Critical level for alevins	$P_c = 1.04 \ln (\text{ATU}_a T'^{-1}) + 0.131 T'$	(27)
P_{50} values for alevins	$P_{50} = 4.81 + 0.127 T'$	(28)

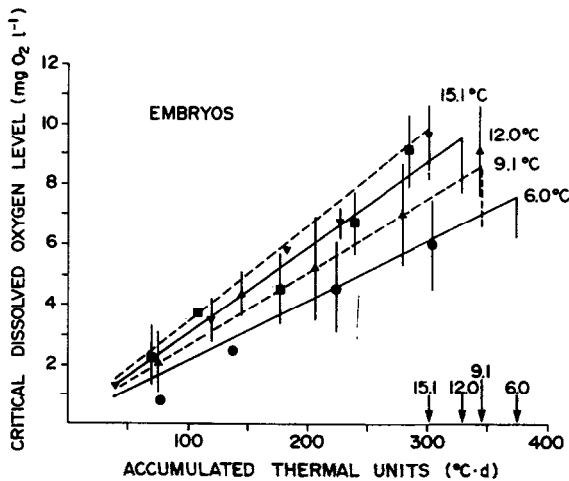


Fig. 6. Critical oxygen levels for steelhead embryos incubated at constant temperatures of 6, 9, 12, and 15°C. Mean values and 95% confidence limits are plotted as functions of accumulated thermal units ($^{\circ}\text{C}\cdot\text{d}$) from fertilization. Curves were fitted using Eqn. 25 relating P_c to incubation period expressed in accumulated thermal units from fertilization and ambient temperature (see text). Arrows indicate accumulated thermal units to 50% hatch.

values for alevins at 6, 9, 12 and 15°C and those predicted using Eqn. 27 are given in Fig. 7.

Salmonid embryos and alevins are normally reared in one of two basic types of incubators: boxes and trays. Oxygen levels in the outflow water of boxes (Atkins cell, Japanese style incubator, gravel box, Bam's box, etc.) can be predicted from the simple equation

$$C = C_i - (\dot{V}\text{O}_2 N Q^{-1}) \quad (29)$$

where C is outflow oxygen level (mg l^{-1}), C_i is inflow oxygen level (mg l^{-1}), $\dot{V}\text{O}_2$ is routine metabolic rate ($\text{mg O}_2 \text{ h}^{-1} \times 10^{-3}$), N is number of individuals in the incubator and Q is flow rate (l h^{-1}). It is somewhat more complicated to predict oxygen levels in tray incubators (e.g., Heath stack) because of reaeration of the water as it falls between trays. However, a mathematical model for predicting the oxygen concentration of the water leaving any particular tray in a series has been developed (W.E. McLean, Quinsam Hatchery, Department of Fisheries and Oceans, personal communication, 1985). The oxygen concentration of water leaving the " n th" tray in a series is given by

$$C = C_s - (\dot{V}\text{O}_2 N Q^{-1} K^{-1}) - (C_s - C_i - (\dot{V}\text{O}_2 N Q^{-1} K^{-1})) e^{-Kn} \quad (30)$$

where C is the oxygen concentration of water leaving the " n th" tray (mg l^{-1}), C_s is the oxygen concentration at 100% air saturation (mg l^{-1}), C_i is the oxygen concentration of the water entering the top tray (mg l^{-1}), K is the reaeration coefficient (tray^{-1}), $\dot{V}\text{O}_2$ is routine metabolic rate ($\text{mg O}_2 \text{ h}^{-1} \times 10^{-3}$), N is number of individuals per tray, Q is the flow rate (l h^{-1}) and n is the

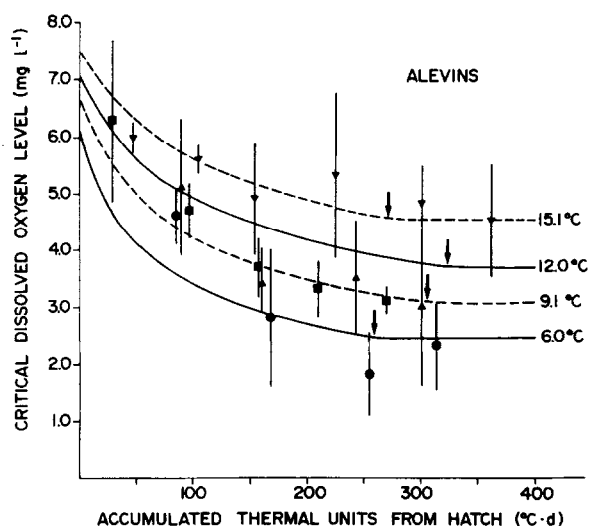


Fig. 7. Critical oxygen levels for steelhead alevins incubated at constant temperatures of 6, 9, 12 and 15°C. Mean values and 95% confidence limits are plotted as functions of accumulated thermal units ($^{\circ}\text{C}\cdot\text{d}$) from hatch. Curves were fitted using Eqn. 27 relating P_c to incubation period expressed in accumulated thermal units from hatch and ambient temperature (see text). Arrows indicate accumulated thermal units to maximum alevin wet weight.

number of the tray in the series. The reaeration coefficient for a standard Heath tray (Heath Techna Corp.) is $0.0555/\text{tray}$ at a flow rate of 12 l min^{-1} (W.E. McLean, Quinsam Hatchery, Department of Fisheries and Oceans, personal communication, 1985).

The first step in estimating dissolved oxygen levels in incubators is to determine which equations to use to predict metabolic rate, P_c and P_{50} . This is accomplished by solving Eqns. 17, 18, 19, 20 and 21 for D , the predicted number of days to the respective stage, using the mean temperature to date ($\bar{T} = \text{ATU d}^{-1}$) as the predictor. If the actual number of days is greater than that predicted, then the embryos or alevins have already passed the stage described by the regression. If the actual number of days is less than that predicted then they have not yet reached that stage. Hence, the segment of development (1–5, Fig. 4) and the equations to use to predict metabolic rate, P_c and P_{50} can be identified. Suppose, for example, steelhead eggs have been incubated for 30 days at a mean temperature of 8.5°C (255 ATU). Completion of epiboly would have occurred after 10 days according to Eqn. 17 but 50% hatch will not occur until day 41.4 according to Eqn. 19. Therefore, the embryo is at a stage of development between completion of epiboly and hatch (segment 2, Fig. 4). Equation 23 (Table 9) should be used to estimate metabolic rate. Equations 25 and 26 (Table 10) should be used to estimate P_c and P_{50} , respectively. Assuming that the actual temperature was 10.0°C on day 30 in this example,

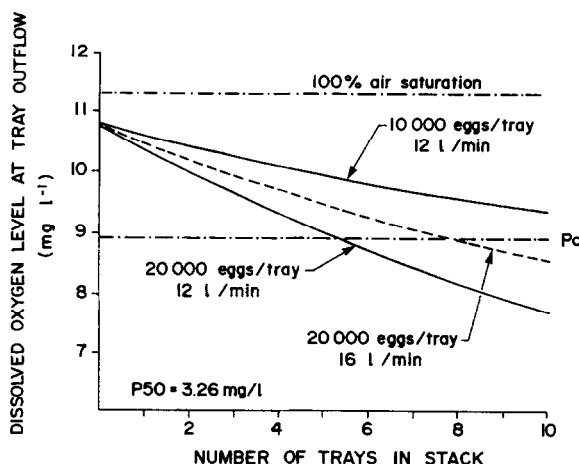


Fig. 8. Predicted dissolved oxygen levels at 10.0°C in the outflow water from the various trays in a Heath stack containing steelhead embryos near hatch (330 ATU). Oxygen concentrations were calculated using Eqn. 30 assuming loading densities of 10 000 and 20 000 eggs per tray, flow rates of 12 and 16 l min⁻¹ and an inflow oxygen concentration of 10.8 mg l⁻¹. Predicted P_c and P_{50} values are shown (see text).

routine metabolic rate would be 6.62 $\mu\text{g O}_2 \text{ h}^{-1}$ according to Eqn. 23, P_c would be 6.8 mg l⁻¹ according to Eqn. 25 and P_{50} would be 2.2 mg l⁻¹ according to Eqn. 26.

The next step in the procedure is to determine the oxygen concentration in the outflow water of the particular incubator in question and to compare that value with the estimated P_c and P_{50} values. Using the example above (255 ATU embryos exposed to an ambient temperature of 10.0°C), the oxygen concentration in the outflow of a box type incubator containing 100 000 eggs would be 9.86 mg l⁻¹ according to Eqn. 29 if the inflow oxygen concentration was 10.8 mg l⁻¹ and the flow rate 12 l min⁻¹. The oxygen level in the outflow of the 10th tray in a Heath stack loaded with 10 000 eggs per tray (total 100 000 eggs) and with the same inflow oxygen level and flow rate would be 10.28 mg l⁻¹ according to Eqn. 30. The higher oxygen level predicted for the Heath stack is a result of reaeration. Dissolved oxygen levels would be above the predicted P_c level (6.8 mg l⁻¹) in both types of incubators. This procedure can be used to predict outflow oxygen levels at any stage of development.

In general, there are two periods when oxygen levels in incubators are likely to be limiting. The first is shortly before hatch when the rate of oxygen consumption, comparatively speaking, is fairly low (Fig. 5) but the critical level is high (Fig. 6). The anticipated oxygen levels in the outflow water of the various trays in a Heath stack loaded at densities of 10 000 and 20 000 late-eyed embryos (330 ATU) per tray is shown in Fig. 8. Water temperature was assumed to be 10.0°C, inflow oxygen concentration 10.8 mg l⁻¹ and flow rates

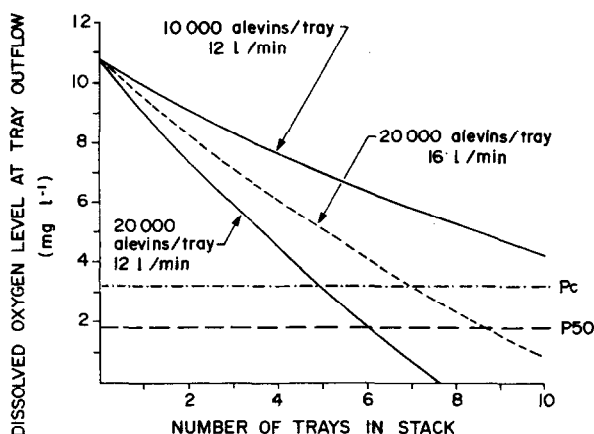


Fig. 9. Predicted dissolved oxygen levels at 10.0°C in the outflow water from the various trays in a Heath stack containing steelhead alevins near ponding (650 ATU). Oxygen concentrations were calculated using Eqn. 30 assuming loading densities of 10 000 and 20 000 alevins per tray, flow rates of 12 and 16 l min^{-1} and an inflow oxygen concentration of 10.8 mg l^{-1} . Predicted P_c and P_{50} values are shown (see text).

12 and 16 l min^{-1} . Oxygen levels are predicted to fall below P_c at a loading density of 20 000 eggs per tray after five trays at a flow rate of 12 l min^{-1} and after eight trays at a flow rate of 16 l min^{-1} .

The other period when oxygen is likely to be limiting is shortly before ponding. Critical levels are relatively low at this time (Fig. 7) but metabolic rates are near their maximum (Fig. 5). Anticipated oxygen levels in the outflow water of various trays in a Heath stack loaded at densities of 10 000 and 20 000 alevins (650 ATU) per tray are shown in Fig. 9. Again, water temperature was assumed to be 10.0°C, inflow oxygen concentration 10.8 mg l^{-1} and flow rates 12 and 16 l min^{-1} . Oxygen levels are predicted to be below P_c after five trays at a loading density of 20 000 alevins per tray and a flow rate of 12 l min^{-1} and below P_{50} after six trays. If flow rate was increased to 16 l min^{-1} , oxygen levels would not drop below P_c until after seven trays and below P_{50} until after nine trays.

Figs. 8 and 9 illustrate how the model can be used to develop an incubation strategy for a hatchery. The hatchery manager can use Eqns. 29 or 30 to predict oxygen levels in the outflow from a box or from the bottom tray in a Heath stack at these two critical periods for given loading densities, flow rates, inflow oxygen levels and temperatures. If predicted oxygen levels in the outflow water fall below P_c , the hatchery manager can alter one or several of the variables in the model, for example flow rate, and recalculate the expected oxygen concentration in the outflow. Alternatively, he might remove the lower trays in a stack, those in which the oxygen level has dropped below P_c , and place them in another unused stack. The manager can arrive at a set of conditions that

will maximize the use of the incubator while maintaining adequate levels of dissolved oxygen by applying this procedure in an iterative fashion.

The model also can be used to predict the oxygen requirements of embryos and alevins at other times in development. The manager can determine the total oxygen demand of an incubation system and predict the flow rate and inflow oxygen level required to meet this demand. Considerable saving in costs associated with water usage could be achieved by matching flow rate to the oxygen requirement at that particular time, as standard flow rates are usually more than adequate during most of the incubation period.

The calculations involved in using the model are fairly complicated and time consuming. Under operating conditions, errors easily could be made if calculations were done manually. The relationships presented in this paper should therefore be incorporated into a microcomputer program. Up-grading of the incubation program currently used by the Federal-Provincial Salmonid Enhancement Program would be the most cost-effective means of achieving this end.

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